Objective
Assess whether borrower risk is better explained by financial strength (income, debt-to-income, installment), historical risk behavior (delinquencies, failed payments, bankruptcies), or loan pricing signals such as interest rate and credit grade. A key question explored was whether interest rate can act as an indicator of risky borrower behavior or whether risk is already embedded in the lender's grading system.

Flow
Conducted exploratory data analysis on borrower financial characteristics, credit history, credit utilization, and loan characteristics. Transformed delinquency-related variables into categorical risk groups, engineered a composite borrower risk score, and analyzed how this constructed risk aligned with income, DTI, installment, interest rates, and credit grades. Findings were further validated using cross-tab analysis and Chi-square testing.

Key Findings
Financial strength variables such as income, DTI, and installment showed weak or inconsistent relationships with borrower risk when analyzed individually, with some patterns appearing counterintuitive. In contrast, interest rates and credit grades showed a clearer alignment with risky borrower behavior. Evidence suggests higher interest rates may reflect a combined assessment of multiple borrower risk factors—particularly debt burden (DTI), credit utilization, and past risk behavior—rather than being driven by income alone. This may explain why interest rates showed a stronger relationship with observed risk behavior than any single financial variable analyzed independently. The findings also suggest loan pricing may act as a practical summary signal of underlying borrower risk.